

# GHSCI output data dictionary

## All potential indicators and output variables

Generated 21 August 2026 using the Global Healthy and Sustainable City Indicators (GHSCI) tool (<https://healthysustainablecities.github.io/>).

This reference describes the output variables that can be produced through the combined GHSCI workflow, including optional GTFS public transport, Earth Engine (large public green space and urban heat vulnerability), cycling accessibility and longitudinal series analyses, across the scales of calculation (sample point, grid, city, custom aggregation areas, and longitudinal series panels). Permutable elements of variable names are represented by parameter placeholders — [destination], [x] (distance threshold) and [network] (cycling route stress measure) — described under Naming conventions and parameters, rather than exhaustively listing every permutation. Regional customisation (e.g. custom destinations, distance thresholds or aggregation areas) yields analogous variables following the same naming patterns; the data dictionary generated alongside each processed study region enumerates the variables actually calculated for that region.

## Study region information

| Indicator             | Variable           | Units | Statistic | Scale      |
|-----------------------|--------------------|-------|-----------|------------|
| Continent             | Continent          |       | category  | city       |
| Country               | Country            |       | category  | city       |
| 2-letter country code | ISO 3166-1 alpha-2 |       | category  | city       |
| Study region          | study_region       |       | category  | city, grid |

## Derived study region statistics

| Indicator                                                                                                      | Variable               | Units                       | Statistic | Scale              |
|----------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------|-----------|--------------------|
| Area (km <sup>2</sup> ; accounting for urban restrictions, if applied)                                         | Area (sqkm)            | km <sup>2</sup>             | sum       | city               |
| Population estimate, as per configured population data source                                                  | Population estimate    | persons                     | sum       | city               |
| Population per km <sup>2</sup>                                                                                 | Population per sqkm    | persons per km <sup>2</sup> | rate      | city               |
| Intersection count (following consolidation based on intersection tolerance parameter in region configuration) | Intersections          | count                       | count     | city               |
| Intersections per km <sup>2</sup>                                                                              | Intersections per sqkm | count per km <sup>2</sup>   | rate      | city               |
| Area (km <sup>2</sup> )                                                                                        | area_sqkm              | km <sup>2</sup>             | sum       | grid, custom areas |
| Population estimate, as per configured population data source                                                  | pop_est                | persons                     | sum       | grid, custom areas |
| Population per km <sup>2</sup>                                                                                 | pop_per_sqkm           | persons per km <sup>2</sup> | rate      | grid, custom areas |
| Intersection count (following consolidation based on intersection tolerance parameter in region configuration) | intersection_count     | count                       | count     | grid, custom areas |
| Intersections per km <sup>2</sup>                                                                              | intersections_per_sqkm | count per km <sup>2</sup>   | rate      | grid, custom areas |

## Analytical statistics

| Indicator                                                                                         | Variable                 | Units | Statistic | Scale              |
|---------------------------------------------------------------------------------------------------|--------------------------|-------|-----------|--------------------|
| Sample points used in this analysis (generated along pedestrian network for populated grid areas) | urban_sample_point_count | count | count     | city, grid         |
| Population grid cell identifier                                                                   | grid_id                  |       | category  | grid, sample point |
| Sample point identifier                                                                           | point_id                 |       | category  | sample point       |
| Count of population grid cells associated with this area                                          | grid_count               | count | count     | custom areas       |

## Naming conventions and parameters

Many indicator variable names are assembled from a fixed stem (identifying the measure) and one or more of the permutable elements below, shown here as placeholders. For example, `pop_pct_access [x]m [destination]_score` and `sp_cycle [network]access [destination]_ [x]m`. Each placeholder, its possible values and defaults are described below; regionally configured values follow the same naming patterns.

### [destination]

The destination whose access is measured. The default destinations are listed below. Walking and cycling access draw on the same destination definitions, declared once in a region's "accessibility" configuration; each pairs a strict and a lenient variant of each core category (food, public open space, public transport) so that composite "all categories reachable" indicators can be derived per variant.

| Destination                                                             | Variable stub            | Category                    |
|-------------------------------------------------------------------------|--------------------------|-----------------------------|
| Fresh food market / supermarket                                         | fresh_food_market        | Food (strict)               |
| Fresh food, pooled (market, supermarket or convenience store)           | fresh_food_pooled        | Food (lenient)              |
| Convenience store                                                       | convenience              | Food                        |
| Large public open space ( $\geq 1.5$ ha)                                | public_open_space_large  | Open space (strict)         |
| Any public open space                                                   | public_open_space_any    | Open space (lenient)        |
| Large public green space ( $\geq 1$ ha; OpenStreetMap and Earth Engine) | large_public_green_space | Open space                  |
| Frequent public transport ( $\leq 20$ min daytime service; GTFS)        | pt_frequent              | Transport (strict, cycling) |
| Public transport, $\leq 20$ min daytime service (GTFS)                  | pt_gtfs_freq_20          | Transport (walking)         |
| Public transport, $\leq 30$ min daytime service (GTFS)                  | pt_gtfs_freq_30          | Transport (walking)         |
| Any public transport stop (GTFS)                                        | pt_gtfs_any              | Transport                   |
| Any public transport stop (OpenStreetMap or custom)                     | pt_osm_any               | Transport                   |
| Any public transport stop (best of GTFS and OpenStreetMap/custom)       | pt_any                   | Transport (lenient)         |
| All categories reachable, strict variants                               | all_strict               | Combined                    |
| All categories reachable, lenient variants                              | all_lenient              | Combined                    |
| Local activity centre (everyday destinations co-located)                | activity_centre_local    | Derived                     |
| Complete activity centre (high-amenity destinations co-located)         | activity_centre_complete | Derived                     |

*Regions can define their own destinations (custom OpenStreetMap tags or supplied datasets), pool categories, and configure additional combined-access sets and activity-centre definitions (a network location whose walkable catchment, default 400 m, contains at least one destination of every required category). These follow the same naming patterns, e.g. `all_<set>_<variant>` for a named combined-access set and `activity_centre_<definition>_<tier>` for a named activity centre. Combined-access composites have access indicators only (no nearest-distance variables).*

### [x]

Distance threshold (m). Distances can be customised to locally policy-relevant values. The core walking access indicators use the configured neighbourhood accessibility threshold (default 500 m). Where a region configures the optional pedestrian accessibility analysis, walking access is additionally evaluated at each configured pedestrian distance, and cycling access at each configured cycling distance (defaults: 500 m, 1000 m, 2000 m and 5000 m), so an access variable exists for each distance.

### [network]

Cycling route stress measure, appearing as an optional prefix on cycling variable names: "lts1\_" for routes composed entirely of lowest-stress Level of Traffic Stress (LTS) 1 links (suitable for all ages and abilities); "safe\_" for fully low-stress routes on LTS 1-2 links (the default headline measure); or no prefix for danger-weighted routing over the full cycling network (higher-stress links usable at a proportionate distance penalty). Which measures are computed is configurable (cycling\_indicators contrasts and measures settings).

## Indicator estimates

### Access (walking)

| Indicator                                                                                                                                                                               | Variable                                | Units     | Statistic  | Scale              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------|------------|--------------------|
| Score (0-1) for walking access within the configured accessibility threshold distance (default 500 m) to the destination                                                                | sp_access_[destination]_score           | score 0-1 | value      | sample point       |
| Distance (m) along the pedestrian network to the nearest destination                                                                                                                    | sp_nearest_node_[destination]           | metres    | value      | sample point       |
| Percentage of population with walking access within [x] m of the destination                                                                                                            | pct_access_[x]m_[destination]_score     | percent   | percentage | grid, custom areas |
| Percentage of population with walking access within [x] m of the destination (population weighted)                                                                                      | pop_pct_access_[x]m_[destination]_score | percent   | percentage | city, custom areas |
| Score (0/1): the destination is reachable within [x] m along the pedestrian network                                                                                                     | sp_walk_access_[destination]_[x]m       | score 0-1 | value      | sample point       |
| Distance (m) along the pedestrian network to the nearest destination                                                                                                                    | sp_walk_nearest_node_[destination]      | metres    | value      | sample point       |
| Percentage of population with walking access within [x] m to the destination                                                                                                            | pct_access_walk_[destination]_[x]m      | percent   | percentage | grid, custom areas |
| Percentage of population with walking access within [x] m to the destination (population weighted)                                                                                      | pop_pct_access_walk_[destination]_[x]m  | percent   | percentage | city, custom areas |
| Average distance (m) along the pedestrian network to the nearest destination                                                                                                            | avg_walk_dist_[destination]             | metres    | mean       | grid, custom areas |
| Average distance (m) along the pedestrian network to the nearest destination (population weighted)                                                                                      | pop_avg_walk_dist_[destination]         | metres    | mean       | city, custom areas |
| Score (0/1): the nearest destination is further than [x] m along the pedestrian network                                                                                                 | sp_walk_beyond_[destination]_[x]m       | score 0-1 | value      | sample point       |
| Percentage of population living further than [x] m along the pedestrian network from the nearest destination                                                                            | pct_beyond_walk_[destination]_[x]m      | percent   | percentage | grid, custom areas |
| Percentage of population living further than [x] m along the pedestrian network from the nearest destination (population weighted)                                                      | pop_pct_beyond_walk_[destination]_[x]m  | percent   | percentage | city, custom areas |
| Number of destinations of the sub-type reachable within [x] m along the pedestrian network                                                                                              | sp_walk_count_[set]_[sub-type]_[x]m     | count     | value      | sample point       |
| Average number of destinations of the sub-type reachable within [x] m along the pedestrian network                                                                                      | avg_count_walk_[set]_[sub-type]_[x]m    | count     | mean       | grid, custom areas |
| Diversity (0-1) of the sub-types reachable within [x] m along the pedestrian network: the Shannon entropy of the counts of each sub-type, normalised by the number of sub-types defined | sp_walk_diversity_[set]_[x]m            | index 0-1 | value      | sample point       |
| Average diversity (0-1) of the sub-types reachable within [x] m along the pedestrian network                                                                                            | avg_diversity_walk_[set]_[x]m           | index 0-1 | mean       | grid, custom areas |
| Share (0-1) of the set's sub-types with at least one destination reachable within [x] m along the pedestrian network                                                                    | sp_walk_richness_[set]_[x]m             | index 0-1 | value      | sample point       |

| Indicator                                                                                                                    | Variable                     | Units     | Statistic | Scale              |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------|-----------|--------------------|
| Average share (0-1) of the set's sub-types with at least one destination reachable within [x] m along the pedestrian network | avg_richness_walk_[set]_[x]m | index 0-1 | mean      | grid, custom areas |

## Walkability

| Indicator                                                                                                                                                                        | Variable                             | Units                       | Statistic | Scale                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------|-----------|--------------------------|
| Average population density of the local walkable neighbourhood (population per km <sup>2</sup> , based on the population grid cells intersecting the sample point neighbourhood) | sp_local_nh_avg_pop_density          | persons per km <sup>2</sup> | value     | sample point             |
| Average street intersection density of the local walkable neighbourhood (intersections per km <sup>2</sup> )                                                                     | sp_local_nh_avg_intersection_density | count per km <sup>2</sup>   | value     | sample point             |
| Daily living score (/3): sum of access scores for fresh food, convenience and public transport destinations                                                                      | sp_daily_living_score                | score 0-3                   | value     | sample point             |
| Walkability index: sum of z-scores of the daily living score, neighbourhood population density and neighbourhood intersection density                                            | sp_walkability_index                 | index (sum of z-scores)     | value     | sample point             |
| Average walkable neighbourhood population density (population per km <sup>2</sup> ) of sample points in this area                                                                | local_nh_population_density          | persons per km <sup>2</sup> | mean      | grid, city, custom areas |
| Average walkable neighbourhood intersection density (intersections per km <sup>2</sup> ) of sample points in this area                                                           | local_nh_intersection_density        | count per km <sup>2</sup>   | mean      | grid, city, custom areas |
| Average daily living score (/3) of sample points in this area                                                                                                                    | local_daily_living                   | score 0-3                   | mean      | grid, city, custom areas |
| Average walkability index of sample points in this area                                                                                                                          | local_walkability                    | index (sum of z-scores)     | mean      | grid, city, custom areas |
| Average walkable neighbourhood population density (population per km <sup>2</sup> ; population weighted)                                                                         | pop_nh_pop_density                   | persons per km <sup>2</sup> | mean      | city, custom areas       |
| Average walkable neighbourhood intersection density (intersections per km <sup>2</sup> ; population weighted)                                                                    | pop_nh_intersection_density          | count per km <sup>2</sup>   | mean      | city, custom areas       |
| Average daily living score (/3; population weighted)                                                                                                                             | pop_daily_living                     | score 0-3                   | mean      | city, custom areas       |
| Average walkability index (population weighted)                                                                                                                                  | pop_walkability                      | index (sum of z-scores)     | mean      | city, custom areas       |

## Cycling accessibility

| Indicator                                                                                                                              | Variable                                         | Units     | Statistic  | Scale              |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------|------------|--------------------|
| Score (0/1): the destination is reachable within [x] m by cycling, routed using the [network] measure                                  | sp_cycle_[network]access_[destination]_[x]m      | score 0-1 | value      | sample point       |
| Distance (m) by cycling to the nearest destination, routed using the [network] measure                                                 | sp_cycle_[network]nearest_node_[destination]     | metres    | value      | sample point       |
| Percentage of population with cycling access within [x] m to the destination, routed using the [network] measure                       | pct_access_cycle_[network][destination]_[x]m     | percent   | percentage | grid, custom areas |
| Percentage of population with cycling access within [x] m to the destination, routed using the [network] measure (population weighted) | pop_pct_access_cycle_[network][destination]_[x]m | percent   | percentage | city, custom areas |

| Indicator                                                                                                            | Variable                                      | Units  | Statistic | Scale              |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------|-----------|--------------------|
| Average distance (m) by cycling to the nearest destination, routed using the [network] measure                       | avg_cycle_dist_<br>[network][destination]     | metres | mean      | grid, custom areas |
| Average distance (m) by cycling to the nearest destination, routed using the [network] measure (population weighted) | pop_avg_cycle_dist_<br>[network][destination] | metres | mean      | city, custom areas |

### Urban heat vulnerability

Derived from Earth Engine raster analyses on a 1 km grid by default, following Turner et al. (2025), "Development and validation of the Global Urban Heat Vulnerability Index (GUHVI)", *Urban Climate* 64: 102716. Inputs are aligned so that higher values indicate greater heat vulnerability. Each measure below is sampled at sample points (sp\_ prefix), averaged for each grid cell or custom area (no prefix), and population weighted for the city and custom areas (pop\_ prefix). Sub-indicators are listed beneath the composite index they contribute to.

| Indicator                                                                                                                                                                                                                      | Variable                                   | Units                       | Statistic  | Scale                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------|------------|----------------------------------------|
| Heat Exposure Index (HEI): the heat exposure sub-index, derived from land surface temperature                                                                                                                                  | urban_heat_exposure_index                  | index 0-100                 | mean       | sample point, grid, city, custom areas |
| Land surface temperature (°C): the temperature of the Earth's surface, indicating the lived experience of heat (Landsat 8 surface reflectance, hottest third of the year)                                                      | urban_heat_land_surface_temp_<br>c         | degrees Celsius             | mean       | sample point, grid, city, custom areas |
| Heat Sensitivity Index (HSI): the equally weighted heat sensitivity sub-index of land surface albedo, NDVI, NDBI, local climate zone, population density and vulnerable population                                             | urban_heat_sensitivity_index               | index 0-100                 | mean       | sample point, grid, city, custom areas |
| Land surface albedo: solar reflectance, where higher reflectance implies less surface heat retention (inverted, so that higher values indicate greater heat vulnerability)                                                     | urban_heat_land_surface_albedo             | proportion 0-1              | mean       | sample point, grid, city, custom areas |
| Normalised Difference Vegetation Index (NDVI): a spectral index of vegetation health and greenness (inverted, so that higher values indicate greater heat vulnerability)                                                       | urban_heat_ndvi                            | index -1 to 1               | mean       | sample point, grid, city, custom areas |
| Normalised Difference Built-up Index (NDBI): a spectral index of impervious surfaces such as concrete and asphalt                                                                                                              | urban_heat_ndbi                            | index -1 to 1               | mean       | sample point, grid, city, custom areas |
| Local Climate Zone: a standardised classification of land cover typologies for urban heat island studies, ordered from least heat retaining (water) to most heat retaining (bare rock or paved)                                | urban_heat_local_climate_zone_<br>ordered  | ordered class               | category   | sample point, grid, city, custom areas |
| Population density (persons per km <sup>2</sup> ), from the GHS-POP global human settlement population grid                                                                                                                    | urban_heat_population_density_<br>per_sqkm | persons per km <sup>2</sup> | mean       | sample point, grid, city, custom areas |
| Vulnerable population (%): those most at risk from heat on the basis of age, being the combined proportion of the population aged 0-4 and 65+ years                                                                            | urban_heat_vulnerable_pop_pct              | percent                     | percentage | sample point, grid, city, custom areas |
| Adaptive Capability Index (ACI): the equally weighted adaptive capability sub-index of child dependency ratio, subnational Human Development Index and infant mortality rate; higher values indicate lower adaptive capability | urban_heat_adaptive_capability_<br>index   | index 0-100                 | mean       | sample point, grid, city, custom areas |
| Child dependency ratio: children (aged 0-14 years) relative to the working age population (15-64 years), indicating economic reliance                                                                                          | urban_heat_child_dependency_<br>ratio      | ratio                       | mean       | sample point, grid, city, custom areas |

| Indicator                                                                                                                                                                   | Variable                                     | Units                        | Statistic  | Scale                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------|------------|----------------------------------------|
| Subnational Human Development Index: human well-being assessed through education, health and standard of living                                                             | urban_heat_subnational_hdi                   | index 0-100                  | mean       | sample point, grid, city, custom areas |
| Infant mortality rate: deaths of children under 1 year of age per 1000 live births, a key indicator of population health                                                    | urban_heat_infant_mortality_rate             | deaths per 1,000 live births | mean       | sample point, grid, city, custom areas |
| Global Urban Heat Vulnerability Index (GUHVI): the equally weighted composite of the Heat Exposure, Heat Sensitivity and Adaptive Capability indices, (HEI + HSI + ACI) / 3 | urban_heat_guhvi                             | index 0-100                  | mean       | sample point, grid, city, custom areas |
| Global Urban Heat Vulnerability Index (GUHVI) class, from 1 (least vulnerable) to 5 (most vulnerable)                                                                       | urban_heat_guhvi_class                       | class 1-5                    | category   | sample point, grid, city, custom areas |
| Percentage of population in Global Urban Heat Vulnerability Index (GUHVI) class 5, the most vulnerable class                                                                | pct_urban_heat_guhvi_class_5_most_vulnerable | percent                      | percentage | grid, city, custom areas               |

## Longitudinal series outputs

*Work in progress (July 2026)*

| Indicator                                                                                                                                                                                          | Variable             | Units | Statistic | Scale               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------|-----------|---------------------|
| Timepoint label of the observation (e.g. year)                                                                                                                                                     | timepoint            |       |           | longitudinal series |
| Output variable name of the indicator observed (see the indicator entries of this data dictionary)                                                                                                 | indicator            |       |           | longitudinal series |
| Value of the indicator at this timepoint for this unit                                                                                                                                             | value                |       |           | longitudinal series |
| Custom area identifier (area panels)                                                                                                                                                               | area_id              |       |           | longitudinal series |
| Unit identifier (generic panels)                                                                                                                                                                   | unit_id              |       |           | longitudinal series |
| Baseline timepoint label of the comparison pair                                                                                                                                                    | t0                   |       |           | longitudinal series |
| Follow-up timepoint label of the comparison pair                                                                                                                                                   | t1                   |       |           | longitudinal series |
| Indicator value at the baseline timepoint                                                                                                                                                          | value_t0             |       |           | longitudinal series |
| Indicator value at the follow-up timepoint                                                                                                                                                         | value_t1             |       |           | longitudinal series |
| Change metric: 'pp_change' (percentage point change; used for bounded percentage indicators), 'diff' (absolute difference) or 'pct_change' (relative change, %, masked where the baseline is zero) | metric               |       |           | longitudinal series |
| Change in the indicator between t0 and t1, as per the change metric                                                                                                                                | change               |       |           | longitudinal series |
| Whether a policy review checklist was configured and scored for this timepoint                                                                                                                     | assessed             |       |           | longitudinal series |
| Policy review checklist version                                                                                                                                                                    | checklist_version    |       |           | longitudinal series |
| Count of policies identified as present in the policy review                                                                                                                                       | presence_numerator   |       |           | longitudinal series |
| Number of policies assessed for presence in the policy review                                                                                                                                      | presence_denominator |       |           | longitudinal series |
| Policy presence score (%)                                                                                                                                                                          | presence_pct         |       |           | longitudinal series |
| Sum of policy quality scores in the policy review                                                                                                                                                  | quality_numerator    |       |           | longitudinal series |
| Maximum possible sum of policy quality scores in the policy review                                                                                                                                 | quality_denominator  |       |           | longitudinal series |
| Policy quality score (%)                                                                                                                                                                           | quality_pct          |       |           | longitudinal series |